Xi Li

Academic Curriculum Vitae — Editorial Application

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Research Profile

My research develops machine-learning methods and evaluation resources at the intersection of **computational pathology**, **spatial and single-cell genomics**, and **multimodal foundation models**. I study how tissue morphology relates to molecular state, how spatial context can be modeled across cells and sections, and how biomedical models should be evaluated under patient, site, assay, and organ shifts. My work spans dataset and benchmark construction, multimodal representation learning, graph learning, generative modeling, and reliability analysis.

Editorial Expertise and Scope

Primary areas	Spatial transcriptomics; single-cell genomics; computational pathology; digital histology; multimodal biomedical AI
Methods	Foundation models; mixture-of-experts; transformers; graph neural networks; diffusion models; representation learning; retrieval and domain generalization
Evaluation priorities	Patient-level and study-level separation; leakage control; data provenance; context-aware reporting; reproducibility; clinically and biologically meaningful validation
Interdisciplinary scope	Machine learning, bioinformatics, computational genomics, medical image analysis, and biomedical data resources

Education

University of California, Irvine — **Irvine, CA**
Ph.D. in Computer Science — *Sep 2023 – Present*

Tsinghua University — **Beijing, China**
M.S. in Computer Science and Technology — *Sep 2019 – Jun 2022*

University of Science and Technology of China — **Hefei, China**
B.S. in Information Security — *Sep 2015 – Jun 2019*

Research Appointments

University of California, Irvine — **Irvine, CA**
Ph.D. Student Researcher — *Sep 2023 – Present*

- Lead research on histology-to-gene prediction, spatial-transcriptomics super-resolution, multimodal biomedical resources, and reliability of biomedical foundation models.
- Developed **STMoE**, a multi-scale, organ-routed framework for full-gene single-cell prediction from whole-slide histology; accepted at CIKM 2026.
- Curated **CellST-24M** and **sMMC-22M**, large cell-aligned histology–transcriptomics resources supporting cross-patient, cross-platform, multi-organ, and context-aware evaluation.
- Developed methods for spatial super-resolution, multi-slice integration, missing-region completion, context-aware foundation-model evaluation, and spatial perturbation-field inference.

Eastern Institute of Technology & Tsinghua University — **China**
Research Assistant — *Jun 2022 – Jun 2023*

- Conducted computer-vision and multimodal-learning research, connecting experimental prototypes with academic and industry collaborators.

Multimedia Lab, Tsinghua University — **Beijing, China**
Researcher, AI Broadcast Production — *Oct 2021 – Sep 2022*

- Developed and deployed an audio–visual cross-attention system for automated multi-camera production at China's National Centre for the Performing Arts.

Peer-Reviewed and Accepted Publications

1. **Xi Li**, Yaqi Hu, Ziheng Duan, Chaoyang Wang, and Jing Zhang. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM)*, 2026. **Accepted**. doi:10.1145/3799682.3841080.

2. **Xi Li**, Yaqi Hu, Ziheng Duan, Diptanshu Sikdar, Yang Liu, and Jing Zhang. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted as a poster**. Conference page.

3. Ziheng Duan, Tong Wu, **Xi Li**, Simon Dongbo Sun, Chaoyang Wang, Junhao Liu, Martin Renqiang Min, and Jing Zhang. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted as a poster**. OpenReview.

4. Ziheng Duan*, **Xi Li***, Zhiqing Xiao, Rex Ying, and Jing Zhang. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM)*, pp. 625–634, 2025. *Equal contribution. doi:10.1145/3746252.3761240.

5. Ziheng Duan, **Xi Li**, Zhuoyang Zhang, James Song, and Jing Zhang. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." In *IEEE International Conference on Image Processing (ICIP)*, pp. 19–24, 2025. doi:10.1109/ICIP55913.2025.11084277.

6. **Xi Li**, Ziheng Duan, Yi Dai, Siwei Xu, and Jing Zhang. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." In *Proceedings of the 35th British Machine Vision Conference (BMVC)*, 2024. Paper page.

Manuscripts Under Review

1. Liyun Lin, Shuozhen Bao, **Xi Li**, Isaiah Yim, Dejiang Wang, Gaocai Li, Xiangjun Di, Shujie He, Junyi Xu, Kathryn Miller-Jensen, Jing Zhang, and Rong Fan. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." *Under review at Nature Communications.*

2. **Xi Li**, Ziheng Duan, Yi Dai, Yang Liu, and Jing Zhang. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." *Under review at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2027.

3. **Xi Li**, Ziheng Duan, Simon Dongbo Sun, Martin Renqiang Min, Yang Liu, and Jing Zhang. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." *Under review at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2027. OpenReview.

4. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." *Under review at Vascular Pharmacology.* Co-authors include Fu Gao and **Xi Li**; corresponding author: Yang Liu. Full author list pending confirmation.

5. Fu Gao, Ian Mason, Mingze Dong, Yao Lu, Di Zhang, Xing Lou, **Xi Li**, Irbaz Hameed, Mingyu Yang, Mei Zhong, Markus Krane, Giovanni Ferrari, George Tellides, Yang Liu, Rong Fan, and Arnar Geirsson. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." *Under review at Circulation; editorial consideration.*

Selected Research Contributions

- **Histology-to-molecular prediction:** Built cell-level and full-gene prediction systems that combine pathology foundation models, multi-scale image context, contrastive alignment, and organ-specialized retrieval.
- **Biomedical datasets and benchmarks:** Constructed multimodal resources spanning tens of millions of aligned cellular records and designed evaluation splits that expose patient, study, platform, and context shifts.
- **Spatial modeling:** Developed methods for arbitrary-scale ST super-resolution, multi-slice integration, missing-region completion, and cell-resolved spatial perturbation-field prediction.
- **Foundation-model reliability:** Developed support-aware evaluation of average and worst-context performance, emphasizing identifiable comparisons, grouped validation, and explicit empirical boundaries.

Teaching and Mentoring

- Graduate Teaching Assistant, University of California, Irvine — three academic terms.
- Research mentor for **Omni-ST: Instruction-Driven Any-to-Any Multimodal Modeling for Spatial Transcriptomics**, UCSC Open Source Research Experience, 2026.
- Mentor junior researchers in multimodal learning, spatial-transcriptomics analysis, reproducible experimentation, and scientific communication.

Selected Professional Experience

ByteDance AI Lab	AI Product Manager Intern; generative-imaging platform and AI-assisted creative tools	2021
ByteDance Feishu/Lark	—Data Product Intern; enterprise search, retrieval, and relevance analysis	2020
Nolibox	Partner and Co-founder; automatic logo generation and font-design tools	2020